

VIGNESHWARAN S

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Portfolio: <https://vigneshwaran-dev.vercel.app>

PROFESSIONAL SUMMARY

Passionate Full-Stack Software Engineer specializing in building scalable, AI-powered, and production-ready web applications using React, TypeScript, FastAPI, Python, PostgreSQL, and modern cloud technologies. Strong foundation in Data Structures, Algorithms, REST APIs, database design, and software engineering principles. Built multiple end-to-end projects, including an AI Developer Assistant CLI integrating Google's Gemini API and a modern portfolio website. Earned HackerRank 5-Star badges in Problem Solving, Python, SQL, Java, and C++, demonstrating strong problem-solving and programming skills. Seeking Software Engineer Internship and Full-Time opportunities to contribute to innovative engineering teams.

EDUCATION

B.Tech – Information Technology

Expected May 2027

SRM Institute of Science and Technology, Trichy, Tamil Nadu | CGPA: 7.06 / 10

TECHNICAL SKILLS

Languages: Python, Java, C, C++ **Frontend:** React, TypeScript, HTML5, CSS3, Tailwind CSS, Vite **Backend:** FastAPI, Flask, REST APIs
Databases: PostgreSQL, Firebase, Supabase **Cloud & DevOps:** Docker, GitHub Actions, Google Cloud Run, Vercel **AI & APIs:** OpenAI API, Gemini API **Tools:** Git, GitHub, VS Code, Postman, Google Colab

INTERNSHIP EXPERIENCE

Full Stack Development Intern — VDart Academy

June 2025 – July 2025

- Engineered reusable React and FastAPI components for production-grade features, aligning with industry coding standards and Agile delivery cycles.
- Designed and integrated RESTful APIs with structured relational databases, enabling reliable, scalable data access across services.
- Collaborated with senior engineers through code reviews and sprint planning, strengthening engineering best practices and delivery velocity.

PROJECTS

Smart Waste Collection System Using IoT and AI | Python • Flask • Firebase • ESP8266 • HC-SR04 • Google Maps API | [GitHub](#)

- Designed an IoT-based monitoring system using ESP8266 microcontrollers and ultrasonic sensors, streaming real-time bin-level data to Firebase for scalable cloud storage.
- Built a Flask-powered dashboard for live visualization and automated alerting, enabling proactive, data-driven waste management decisions.
- Optimized collection routes with Google Maps API-driven AI logic, reducing unnecessary trips and improving operational efficiency.

AI Developer Assistant CLI | Python • Google Gemini API • Markdown • CLI • [GitHub](#)

- Developed a modular AI-powered command-line developer assistant that integrates Google's Gemini API to help developers generate code, explain concepts, debug programs, and create technical documentation.
- Designed a clean, extensible architecture with reusable Python modules, configuration management, Markdown rendering, conversation history, and interactive CLI workflows for an enhanced developer experience.
- Implemented secure environment-based API key management, customizable prompts, session persistence, and production-ready project structure following software engineering best practices.

Portfolio Website | React • TypeScript • Tailwind CSS • Framer Motion • Vite | [GitHub](#)

- Developed a responsive personal portfolio showcasing projects, technical skills, certifications, achievements, and resume.
- Built reusable React components with Tailwind CSS and Framer Motion to deliver smooth animations and a modern user experience.
- Optimized performance using Vite and integrated Resume, GitHub, LinkedIn, LeetCode, and CodeChef profile links.

CERTIFICATIONS

- [Getting Started with AI](#) — IBM SkillsBuild
- [Programming with JavaScript](#) — Meta
- [Databases and SQL for Data Science](#) — IBM
- [Artificial Intelligence for AI](#) — Infosys Springboard

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- Solved 300+ [LeetCode](#) problems across Data Structures, Algorithms, Dynamic Programming, Graphs, Trees, Binary Search, and Backtracking.
- [CodeChef](#): Achieved 3-Star rating (Division 2, 1616) with a DSA Rating of 2023.
- [HackerRank](#): Earned 5-Star badges in Problem Solving, Python, SQL, Java, and C++; earned 3-Star badges in C and React.
- Recognized at TEXPERIA 2026 Hackathon for technical project presentation and innovation.
- Completed 6+ industry certifications in AI, Cloud, SQL, JavaScript, and NLP from IBM, Meta, Google, and Infosys.